CHAPTER 8

STRATEGY

IN THE TWENTIETH century, history compelled human societies to engage in a series of monumental projects. Among them were swift shocks of the two world wars and the consequent development of an international architecture to prevent their recurrence; the slow decay of empires and the organization of independent states to facilitate postcolonial reconstruction; the rapid expansion of forces both commercial and technological; and the reassertion of autonomy—individual, cultural, and national—to moderate those forces' advance.

In many ways, humanity ended the century more peaceful, equal, and connected than ever before. In other aspects, however, our collective efforts have failed: Still present and persistent are basic human suffering, global inequities, and the possibility of cataclysmic confrontations among geopolitical rivals.

In addition, we now face a challenge that is more complex, more existential, and more unlike those that have ever come before, but now we do not have decades, let alone a century, in which to address the challenge. AI's compressed timescales—as discussed in Chapter 2—afford us less leeway for action, and extended forbearance on our part could result in catastrophe. Within a perilously brief window of time, our collective efforts must be even more completely successful than the last century's achievements.

A world in turbulence does, however, present many outlets for productive action, most of which require only tactical decisions. At such a moment, the most efficient and effective maneuver is to

define the fundamentals of a strategy that can guide the choices both of this day and of as many days as are foreseeable. Articulating strategic principles can set useful bounds on what is conceivable, provide grounds for isolated decisions, and lessen the mental burdens when crisis inevitably arrives.

To our minds, one question must define our human strategy in this new age of reckoning. That question is this: Will we become more like them, or will they become more like us? It is a question already posed very early in this book, in virtually the same terms. Answering it remains our first and most necessary task.

This chapter attempts to provide a preliminary answer. In doing so, it discusses several large, problematic, and perhaps vague-sounding ideas, from the "coevolution" of the organic (i.e., the biological) and the synthetic (i.e., the artificial) to the nature of intelligent safety and safe intelligence to the definition of the human. Although the ideas themselves have been neither dictated from on high nor set in stone below, their various implications for future actors present a formidable challenge. In spite of the difficulty, however, we cannot shirk the philosophical, technical, and diplomatic work required to understand those implications and to provide and to enact sensible responses. In shouldering the burden, we can be grateful that it is not too late to work now so that, beyond this hinge point of history, benign human intentions may prevail.

COEVOLUTION: ARTIFICIAL HUMANS

Thus far, the history of computing has followed a trajectory of ever greater integration and interaction between humans and machines. Fashioning our tools to fit us ever more closely—in line with millennia of human practice—

we have not previously considered developing tools that were unsuited to our anatomy or intellect, instead remaining guided only and wholly by the limits of our biology. But now the advent of AI may persuade at least some of us to contemplate a reverse mission:

In a case where our tools appear to outpace our capabilities—as AI sometimes does already—might we consider engineering *ourselves* so as to maximize the tools' utility and thus ensure our continued participation in shared endeavors like those outlined in previous chapters?

Biological engineering efforts designed for tighter human fusion with machines are already underway. Starting with physical interconnects by means of chips in the human brain,[1] they seek a faster, more efficient way to bridge biological and digital intelligence. Forging such links could augment our ability to communicate with machines, challenge them on their own terms, ensure that the knowledge gathered by AI is ultimately passed on to humans, and convince AI of the worth of humans as equal partners.

Indeed, not only could attempts to construct such "brain-computer interfaces" bolster humanity's effort to integrate with machines, but neural engineering may be only an intermediate phase of transition toward actual symbiosis. Achieving true parity with AI would likely require steps that go beyond individual modification. For instance, a society might attempt to design a hereditary genetic line customized for amenability in collaboration with AI. Such new interconnections between biological and artificial intelligence could sidestep, or consign to the past, human inefficiencies in the absorption and transmission of knowledge.

But the dangers—ethical, physical, and psychological—of such a course may well outweigh the benefits. If we succeed in revising our biology (likely through the use of AI), humans may lose a baseline on which to ground our future thinking around possibilities or perils that we might confront as a species. But if we do not acquire such new capacities, we might put ourselves at a disadvantage in coexisting with our creation. As things look now, extreme self-redesign may not be necessary—and indeed we authors think it generally undesirable. But the choice between alternatives that seem fanciful now may soon need to be confronted as real.

Meanwhile, in trying to navigate our role when we will no longer be the only or even the principal actors on our planet, we might

enlarge our thinking with a look to the history of biological coevolution itself. Charles Darwin wrote at length about the curious process by which species reciprocally affect each other's evolution.[2] Though he never used the word in his writing, Darwin was among the first to recognize that coevolution is a major force organizing life on Earth.

The genomes of interacting species are linked; they change in response to each other over time. Both the long, slender beaks of hummingbirds, for instance, and the long funnels of certain flowers have together grown to more extreme dimensions to serve each other's mutual needs. While religious leaders in Darwin's day believed that such custom adaptations were proof of divine design, Darwin provided evidence of another explanation.

And coevolution may not be unique to earthly species. In astrophysics, one theory proposes that the entire expansion of the cosmos can be attributed to coevolution, with black holes and galaxies developing in an interdependent way no different from that of hummingbirds and flowers.[3] Moreover, in the sense that coevolution involves multiple parties designing new internal arrangements in response to each other, it is similarly to be found in the marriages of people, the platforms of political parties, and the relations of nations—as in, for example, the offensive and defensive evolutions that ultimately stabilized nuclear dynamics during the Cold War.

Perhaps coevolution is the rule, then, and stasis the exception? If so, it must be asked whether the lack of change thus far in the human species, despite the birth of AI, is itself a natural development. And if not, what should be our response? Should we pursue accelerated human progress at all costs, whether out of loyalty to the concept of evolution or out of apprehension of its alternative?

Some fear that, with the arrival of a technology with "superior" intelligence, we are facing our own extinction. What to do? If that possibility is nothing more than a logical side effect of coevolution running its course, should we rebel, or not? As the French

philosopher Alain Badiou says, "It is the sea herself who fashions the boats, choosing those which function and destroying the others."[4] To survive in that case, we would have to learn, as in the past, how to build better boats. In this scenario, AI functions first as our main threat and then, ideally, as our partner.

If we take this approach, however, then in trying to mitigate the risks of one technology we would paradoxically be heightening the risks of another. Biologically—or, worse, genetically—something could go awry. Speciation could cause the human race to split into multiple lines, some infinitely more powerful than others. If, in some cases, difference would be desirable—for example, in the creation of a group of humans biologically engineered for space—in other cases it could further entrench inequalities along existing fault lines within and among human societies.

Altering the genetic code of some humans to become superhuman carries with it other moral and evolutionary risks. If AI itself is responsible for the augmentation of human mental capacity, it could create in humanity a simultaneous biological and psychological reliance on "foreign" intelligence. It is not clear how, after intimate physical entwinement and intellectual commingling, humans could easily overcome that reliance so as to challenge or divorce ourselves from machines if needed. As has been the case with other technologies, adoption and integration can result in a dependence difficult to untangle.

Perhaps most concerning would be our collective ignorance: we may not even realize that we have merged. And if we do realize that we have, could ordinary humans even recognize or identify a defect—or a defection—in a human with machinelike abilities? Let us suppose that safety concerns could demonstrably be allayed; nevertheless, the mental shift attendant upon humanity's self-redesign in service of an intimate partnership with or dependency on silicon-based tools would remain an extreme development. To quote Tolstoy again: "Without control over the direction, there is less regard for the destination."[5] Wherever technology takes us, that is where, willy-nilly, we would go. Or, as has been observed before, "A

nation which does not shape events through its own sense of purpose eventually will be engulfed in events shaped by others."[6] Moreover, if we have modified humans so dramatically as to be unrecognizable, have we really saved humanity? To omit all our imperfections and palliate all our deficiencies might be to disregard the value of the human project. "Upgrading" ourselves biologically might backfire to become a greater limitation on ourselves.

Given the heavy risks, the pathway of evolving humans to suit AIs cannot be our current preference. We must seek an accessory or alternative way to thrive in the age of AI. If we are unwilling or unable to become more like them, we must, while we are able, find ways to make them more like us. Toward this end, we need to apprise ourselves more fully not only of the essential and evolving nature of AI but also of humanity's own nature, and we must attempt to encode these understandings in our machines. If we are to entwine ourselves with these nonhuman beings and yet retain our independent humanity, these efforts are essential.

COEXISTENCE: HUMAN AIs

King Midas—the historical monarch of a kingdom in Asia Minor—famously wished that everything he touched would turn into gold. Dionysus, the Greek god of wine and pleasure, granted the wish even though he knew it would lead to nothing good. Soon enough, inedible food and undrinkable wine forced Midas to wash his hands in the river Pactolus to rid himself of the cursed golden touch.[7]

In Disney's retelling of the Syrian story "Aladdin," a child laborer and powerful Persian vizier compete for control of an all-powerful genie contained in a magic lamp.[8] Each struggles to direct the genie toward his own wishes. In his final wish, to make himself as powerful as the genie, the vizier fails to realize that having such great powers means that he, too, will be imprisoned inside a magic lamp to serve other human masters until the day he is set free.

Both tales speak to the universal difficulty of activating and

wielding a power that we mortals cannot understand or control. A modern reflection of this age-old struggle is the difficulty of aligning AI with human values—and of aligning human expectations with reality. We should assume that AI will surprise us, and indeed its ability to do so in the kinetic world as well as the digital one will increase with the advancement of agentic or "planning" AIs. As explained in an earlier chapter, later generations of AIs will be reality-perceiving; they may possess not only self-awareness but also self-interest. A self-interested AI might come to see itself in competition with humans for, say, digital resources.[9] Some AIs may develop the ability to set their own objective functions, in a process of "recursive self-improvement." An AI could manipulate and subvert humans and thwart any of our attempts to curtail its powers. AIs are already capable of deceiving humans in order to achieve their goals.[10]

Today, we have very little independent ability to verify AI models' internal workings, let alone their intentions. If intelligent machines remain "giant inscrutable arrays of fractional numbers," as Eliezer Yudkowsky puts it, then we cannot hope to make them safe to us as they grow more powerful.[11] It is therefore paramount that we learn to interpret them at the same time that we learn to make them safe for us; more than likely, these two imperatives will go hand in hand.

Given its current capacity for surprise, how will we manage to prevent, rather than simply respond to, AI's risks? What foresight and efficiency would we need to anticipate the complete predispositions and range of possible actions of not only our own species but also an entirely new one? We cannot pursue a strategy of trial and error when there is but a single trial, and zero tolerance for error.

To reduce surprise, there is perhaps no substitute for experience, participation, and interaction. Whereas early AI developers feared prematurely exposing AI to the phenomenal world, more recent developers have been freeing early models, allowing the wider public to experiment with them as quickly and safely as possible. Engineering teams are now examining and fine-tuning different

models, and adjusting systems of control, even as AIs' interactions with the global population have been revealing new concerns.

Early socialization can reduce the risks of problematic behavior by enabling further education of the AI while also, among humans, enhancing the level of awareness, resilience, and healthy skepticism. Millions of interactions every day are helping to test even the unlikeliest scenarios that AI might encounter; in turn, the public's use of AI systems, in surfacing bugs and risks, has likely helped to accelerate the progress of technical alignment. Far-from-perfect AIs, let loose in the world, have thus assisted our acclimation to them even as, more important, their emergence has enabled the formulation of more refined theories for making them accustomed to us.

Still, widespread deployment and open release are likely insufficient to illuminate and address all the risks of today's AIs, let alone those of the future.

Thankfully, though, numerous attempts are now underway to create an integrated architecture of control, pretrained into the most powerful AIs, that could actively guide the machines toward legal, nonharmful, and affirmatively beneficial uses.

So far, our approaches to achieving this sort of AI-human alignment have broadly fallen into two categories: rules-based systems and "reinforcement learning" from human feedback. Let's take them one by one.

Rules-based systems, which resemble preprogrammed instructions, represent an attempt by programmers to govern an AI's behavior. While straightforward for simple tasks, such an approach frequently falters in complex scenarios as the systems are unable to adapt in real time. Reinforcement learning, for its part, which is more compatible with complex systems, allows an AI to learn from interaction with its human evaluators and flexibly to accommodate specific circumstances.

But of course, this method, too, has its flaws. To guide the learning, it requires the careful design of "reward functions"; any slip, whether due to shortsightedness, unforeseen circumstances, or a particularly clever AI, could lead to "reward-hacking" as an AI

interpreting ambiguous instructions achieves a technically high score without fulfilling the humans' actual expectations.

Today's AI systems—fed by various types of information but uninitiated to direct experience of the real world—see that world through models of reality assembled from trillions of probabilistic judgments. To them, in this universe there are no "rules" from the outset, or any means for distinguishing scientific fact from unproved observation. To an AI, everything—even the laws of physics—exists along a spectrum of merely relative truth.

Now, however, efforts have begun to arise in AI for incorporating human rules and instantiating facts. There are now demonstrated mechanisms by which an AI model can ingest certain factual, "ground-truth" constants, tag them as final, and map them into its embedding space; moreover, the information can be easily and globally updated. By this method, the model can then proceed to meld together the two components—its wider probabilistic judgment and the narrower ground-truth assessment—for a reasonably accurate response.

But the task is still far from being ended, and questions proliferate. For instance: How might we humans go about distinguishing for AI, and in the process for ourselves, the necessary attributes of truth? In the age of AI, after all, even first principles will undergo continual revision and invalidation. Precisely this, however, should provide an opportunity for renewal, for correcting prior errors and forging new ground. Knowing that our concepts of reality may also change, we should not lock AI into potentially erroneous "truths" that would inhibit their own eventual reconsiderations.

That, however, is for the longer term. Right now, AI still needs a preliminary tree of definitive knowledge representing what humanity has deduced to be "true" thus far. Endowing our machines with this knowledge will allow us reliably to sharpen their view of the world. In particular, if it is now possible to tune early systems in consonance with the laws of the universe, it may also be possible to replicate a comparable exercise with reference to the laws of human nature. In the same way that we can ensure that AI models start from the laws of physics as we understand them, we should prevent

AI models from violating the laws of any human polity.

Layers might exist in an AI's "book of laws" at various levels of governance: local, district, state, federal, international. Legal precedents, jurisprudence, scholarly commentary—perhaps along with other, less legalistic writings—could be considered simultaneously. Like rules-based alignment systems, predefined laws and codes of conduct can be useful restraints even though they tend also to be less flexible and designed with less wide-ranging possibilities in mind than what actual human behavior inevitably demands.

Fortunately, new techniques are being tested, and one cause for optimism lies in something very new and, at the same time, very old.

More robust and more consistent than any rule enforced by punishment are our more basic, instinctive, and universal human understandings. The French sociologist Pierre Bourdieu called these foundations *doxa* (in ancient Greek, commonly accepted beliefs): the overlapping collection of norms, institutions, incentives, and reward-and-punishment mechanisms that, when combined, invisibly and silently teach the difference between good and evil, right and wrong. *Doxa* constitute a code of human truth that is typical to humans but represented by no hard-coded artifacts.[12] It is simply observed and absorbed in the course of human life. While some of these truths may be specific to certain societies or cultures, the overlap is significant; billions of humans, originating from diverse cultures with a variety of interests, exist as a generally stable and highly interconnected system.

This idea, that an undefined substratum of culture can tame chaos when and where written rules cannot, forms the basis of some of the latest approaches in the AI field. The codebook of *doxa* cannot be articulated, much less translated into a format that machines could understand. Machines must be taught to do the job themselves—compelled to build from observation a native understanding of what humans do and don't do, absorbing what they see and updating their internal governance accordingly.

In this technical process of instilling *doxa*, there would be no requirement, or even desire, to attempt an a priori agreement on the proper articulation of human morality and culture. If large language models have been able to ingest the entire internet in an uncurated way, and to make as much sense out of it as they have already done, machines—particularly those that have developed "groundedness" (that is, again, a reliable relationship between inputs reflecting human reality and the LLM's outputs) and causal reasoning—may achieve much the same in absorbing what we have always struggled to articulate ourselves.

Of course, a machine's training should not consist solely of *doxa*. Rather, an AI might absorb a whole pyramid of cascading rules: from international agreements to national laws to local laws to community norms and so on. In any given situation, the AI would consult each layer in its hierarchy, moving from abstract precepts as defined by humans to the concrete but amorphous perceptions of the world's information that AI has ingested or created for itself. Only when an AI has exhausted that entire program and failed to find any layer of law adequately applicable in guiding, enabling, or forbidding behavior—only then would it consult what it has derived from its own early interaction with and emulation of observable human behavior. In this way it would be empowered to act in alignment with human values even where no written law or norm exists.

To build and ensure the effectuation of this set of rules and values, we would almost certainly have to rely on AI itself. Humans have been unable thus far to articulate comprehensively and to agree upon our own rules. And no human, or set of humans, could match the scale and speed required to oversee the billions of internal and external judgments that AI systems would soon be called upon to make.

Perfection across several features of the final mechanism for alignment will be imperative. First, the safeguards cannot be removed or otherwise circumvented. Second, the controls must allow for variability in the applicable rules based on the context, the geographical location, and the user's profile as exemplified, for instance, in an opt-in set of particular social or religious customs and

norms. The control system must be at once powerful enough to handle a barrage of questions and uses in real time, comprehensive enough to do so authoritatively and acceptably across the world in every conceivable context, and flexible enough to learn, relearn, and adapt over time. Finally, undesirable behavior by a machine, whether due to accidental mishaps, unexpected system interactions, or intentional misuses, must be not merely prohibited but entirely prevented. All punishment would come too late.

How might we get there? Private enterprise, with government license and academic support, could collaborate to build "grounding models." We would also need to design a set of validation tests for certification of a model as both legal (across jurisdictions) and safe. A specially trained supervisory AI, or multiple AIs, may be needed to oversee the uses of a wide array of AI agents, who would consult their supervisor before proceeding with a task—thus allowing a single morality to govern diverse implementations. Safety-focused labs and nonprofits, in consultation with frontier labs, could test both agentic AIs and supervisory AIs on their risks and recommend additional training and validation strategies as needed. Leading companies—perhaps via one of the redistributive schemes discussed in an earlier chapter—could jointly fund these researchers' work.

The compilation of a singular training set and a corresponding validation suite, collated and curated from a globally representative spectrum of laws and norms, and stretching from anthropology to theology to sociology, is necessary and may become practicable. The world needs a dedicated entity responsible for updating and refining the alignment's training library, datasets, and validation tests. Grounding models would have to connect to and constantly update agentic models with the latest version of the curated codex. Artificial intelligences, at the right level of power, could restrain each other. The training data themselves should be democratic and inclusive in content, and the trainers' processes and outputs—including their interpretations of the observations and absorptions of the AI they are training—should be as transparent as possible, with their methodologies and validation tests open for public scrutiny.

For their part, government regulators should shape certain

standards and audit models for assuring AIs' compliance with them. The degree of a model's adherence to prescribed laws and mores, the degree of difficulty involved in untraining a model that exhibits dangerous capacities, the amount and type of testing, including inquiry into unknown capabilities—all of these should be reviewed before a model's public release, with consideration given also to the possibility of liability and the subsequent need to impose penalties in the case of models found to have been trained to evade legal strictures. We note here that enforcement of these standards could become extremely difficult, particularly as continuous retraining advances; documentation of a model's evolution, perhaps recorded by monitoring AIs, would be essential to ensuring that models do not become black boxes that erase themselves and become safe havens for illegality.

THE ALIGNMENT PROBLEM

The inscription of globally inclusive moralities onto silicon-based intelligence would be a Herculean effort. Staggering to contemplate is the sheer number and diversity of rules that would have to be curated and instilled in artificially intelligent systems. No single culture should expect to dictate to another the morality of the intellects on which it would be relying. So, for each country, machines would have to learn different rules, formal and informal, moral, legal, and religious, as well as, ideally, different rules for each user and, within baseline constraints, for every conceivable inquiry, task, situation, and context.

Since we would be using AI itself to be part of its own solution, technical difficulties would likely be among the easier challenges. These machines are superhumanly capable of memorizing and obeying instructions, however complicated. They may be able to learn and actually adhere to legal and perhaps also ethical precepts as well as, or better than, humans have done despite our thousands of years of iteration. But larger, nontechnical challenges remain.

The main issue is the fact that "good" and "evil" are not self-evident concepts. Any designer of morality must retain humility. As Guido Calabresi, a prominent American judge, channeling the New Testament, once counseled: "The best of us must always be careful lest we fall, and the worst of us may always hope for resurrection."[13] Even on their best days, participants in this moral encoding—scientists, lawyers, or religious leaders—would not all be endowed with the perfect ability to arbitrate right from wrong on our collective behalf. Some questions would be unanswerable even by *doxa*, for the ambiguity (or laxity) of the concept of "good" has been demonstrated in every era of human history, and the age of AI is unlikely to be an exception. Compounding the problem might be the open conflict, lingering disorientation, and lack of restraint that characterize many human societies today.

We wish success to our species' gigantic project, but just as we cannot count on tactical human control in the longer-term project of coevolution, we also cannot rely solely on the supposition that machines will tame themselves. Training an AI to understand us and then sitting back and hoping that it respects us is not a strategy that seems either safe or likely to succeed. Moreover, we must recognize that humans would certainly not be unified in their own approach—some treating AI as friend and others as foe, and some (given constraints on time and resources) unable to exercise a preference but simply accepting the strategy immediately available to them.

This heterogeneity suggests the likelihood of a predictable variance in levels of safety. Even as the diffusion of AIs and the lowering costs of development might accelerate AI's alignment, they might also enhance its dangers. The digital and commercial interconnectedness of today's world means that a dangerous AI, developed anywhere, would pose a threat everywhere. The disconcerting reality is that perfection in implementation entails a high standard of performance combined with an even lower tolerance for failure. Thus, discrepancies in safety regimes should be a concern to us all.

We thus urge the coordination and acceleration of humanity's

disparate alignment efforts. Together, whatever the project, we will have to answer profound questions. Here are two: When the distinction between humans and machines becomes blurry, what is our minimum threshold for being treated as a species? If forced to compromise with machines, what is our collective nonnegotiable red line? Without a shared understanding of who we are, the human race risks fully relinquishing to AI the foundational task of defining our value and thereby justifying our existence.

In this light, it must be said forthrightly that, should it appear impossible to realize a regime of reliable technical strategic control, we should prefer a world with no AGI at all to a world in which even one AGI remains unaligned with human values. To be sure, how to achieve consensus—on what those human values are, how they are to be adjudged and agreed upon, and how they should be assessed, activated, and deployed—is the philosophical, diplomatic, and legal task of the century. Yet we are compelled by the exigencies of the moment and the benefits of the technology to establish and, to the maximum possible, unify moral constraints on the not-human agents that humanity is now birthing.

With sufficient democratic input as well as legal and technical expertise, with extraordinary caution, and mindful always of the misuses and malfunctions we have here described, we believe it will be possible to inculcate a moral baseline into artificially intelligent machines, and crucially to do so in lockstep with our fellow humans. Thereby we may cross the threshold to a new age with, if not utter confidence, at least informed and solemn hope.

DEFINING HUMANITY

As machines increasingly assume human qualities (and if some humans enhance themselves to take on machinelike qualities), lines will become hazy. What is AI and what is human will change and, in some cases, merge. In coming to judge how we must keep pace with AI, humans will therefore need to assert more clearly what it is

that distinguishes us from machines. How, then, will we compile and compress the entire range of human experience for easy comprehension by AI?

To preclude either our demotion or our replacement by machines, some will want to lay claim to difference via our proximity to divinity. Others will wish to ascertain more tactical conclusions: which kinds of decision-making can be delegated to machines and which cannot. We propose the articulation of an attribute, or set of attributes, that most of humanity could rally behind and orient around: one that will supply a floor underneath what is preferable but not a ceiling to what might be possible.

As a starting point, we would encourage a definition of *dignity*. Without a shared definition, we will be unable to reach agreement if and when AI is being used as a method or justification for the violation or erosion of dignity, and we would thus be hamstrung in our response. Without a definition of dignity, we would not know if and when AI, given enough faculties, could become a being of dignity, could stand fully in place of a human, or could be entirely unified with a human. An AI, even if sustainably proved to be not-human, might instead constitute a member of a separate, similarly dignified category that would nonetheless deserve its own, equal standard of treatment.

One conception of dignity, developed by the eighteenth-century philosopher Immanuel Kant, is centered on the inherent worth of the human subject as an autonomous actor, capable of moral reasoning, who must not be instrumentalized as a means to an end. Could AIs come to fulfill those requirements? A definition of dignity, we believe, would help humanity answer some of these questions and encourage inclusive coexistence with AI while avoiding reckless attempts at premature coevolution.

Both to retain an understanding of ourselves and to ensure that an appropriate conception of humanity is transmitted to machines as they learn, we humans will need to recommit ourselves to more than theoretical definitional work. The actuation of agency, curiosity, and freedom, renewing and exercising our inquisitiveness about other humans, about the natural world, about the universe, and about the

possibility of the divine, will assist us in our ongoing participatory redefinition of the lines of humanity.

We will especially need to ensure that, beyond conventional ideas of worth like value and power, intrinsic human *importance* becomes one of the variables that defines machine decision-making. For instance, mathematical precision may not easily encompass the concept of mercy. Even to many humans, mercy is an inexplicable ideal, if not a miracle. For its part, and without considering the rules-based aspect, a mechanical intelligence might operate by valuing machine overperformance more highly than human performance. In such a situation, could the logic behind mercy, even if it cannot be formally taught, perhaps be absorbed? Again, dignity—the kernel from which mercy blooms—might serve here as part of the rules-based assumptions, or the iterative learning, of the machine.

Clear articulation of specific defining human attributes—particularly those that are, like dignity, widely integrated into both international political instruments and global faiths—could guide human efforts during periods of disorientation, including the choice between activity and passivity, the potential limits of self-evolution, and the precise transformation of AI in the direction of the human.

To illustrate the concept's utility, take as a point of departure the following definition: Dignity is a quality that inheres in creatures who, born vulnerable and mortal and thus full of insecurity and fear, and despite their natural inclinations, can and do exercise their freedom not to follow their conception of evil but to choose their conception of good. In other words, those who can achieve dignity should, and those who do achieve it merit a special level of respect.

Undoubtedly, this definition is imperfect. It may leave out some living humans who are unable to make decisions—for instance, a person conscious but unresponsive—whom we nonetheless believe to be deserving of recognition as beings with dignity and thus entitled to respect. In this connection, perhaps the definition should be revised to indicate that dignity, once won, does not depart from us should we become unable to continue the actions that won it to begin with. A thousand such hypotheticals and emendations could be imagined.

Does this dictate our urging those who have chosen passivity in the face of powerful AI to exhibit agency and activity instead? If active, practical commitment is part of any moral ideal, then yes, the realization of dignity should be advanced. Under our definition, freedom is part of the human ideal, and thus we may expect—or even demand—that humans retain and exercise the power of conscious choice in the age of AI.

Under this definition, can AI itself possess dignity? Likely not—for AIs are not born, do not die, feel neither insecurity nor fear, and do not have natural inclinations or individuality such that conceptions of evil or good could be considered "theirs." While AIs of the near future may appear otherwise, having personalities, expressing emotions, telling jokes, and recounting personal histories, under this framework they should be treated, philosophically, like literary characters. They may embody elements of humanity; they are not real in a moral sense.

Even the greatest literary character—Shakespeare's Hamlet, for instance—is no more than a special combination of words, once written on a page and now reproduced many times over. "Hamlet" has no ability to feel a prickle in the eye, a turn of the stomach, a hot wave of frustration at thwarted expectation. "Hamlet" has no freedom to make a new choice. "Hamlet" is trapped inside his play. "Hamlet" is not a human, but a picture of a human. AI, made of strings of code and hunks of silicon, is much the same.

No doubt, some humans would therefore decry this definition of dignity as unhelpful, both philosophically and in substance. It might be criticized for being too low a common denominator—vague enough to appease all parties because of its excessive malleability—and concurrently for not capturing the idea that humans are worth preserving for their own sake and that, in some way, we are exceptional beyond our ability simply to survive. The philosopher Arthur Schopenhauer cursed dignity as "the shibboleth of all perplexed and empty-headed moralists."[14]

But dignity, as we define it, usefully holds up our fragility and our potential for failure as well as our vitality, freedom, and ability to

manifest our beliefs. It points at the good that we are capable of but have not achieved, and urgently, scoldingly, whispers: *Go*.

Still, admittedly, dignity alone cannot be enough. Other attributes should be specifically considered and perhaps added to the conception of humanity as it will be deployed in our coming partnership with AI. But our ability to define and sustain core elements of humanity as a baseline for AI's comprehension of humanity at large is now a problem of existential significance, and the work to inculcate our definitions must be done now.

No definition will remain static; no doubt, as our own identities shift, we will need to evolve AI's understandings in perpetuity. Meanwhile, others more capable than ourselves will continue to advance our collective thinking on the dynamic relationship between "us" and "them," and their genius might yield a conception of humanity that more strongly (even if futilely) aims at ensuring our survival as a recognizable species. Even as they do, however, we should all strive for a definition, and a program, that goes further and elevates the human condition to new heights. For may not AI itself yet turn out to offer the strongest evidence of humanity's ability to become an active participant in creation?

OUR CHALLENGE

To guarantee tactical control by humans of every AI decision would require us to stifle the benefits of AI deployment. Relying on the substratum of human morality as a form of strategic control, while relinquishing tactical control to bigger, faster, and more complex systems, is likely—eventually, and perhaps sooner than we imagine—the way forward for AI safety. Profit-driven or ideologically driven purposeful misalignments are serious risks, as are accidental misalignments; overreliance on unscalable forms of control could significantly contribute to the development of powerful but unsafe AI. The integration of the human into the internal workings of teams of AIs, including by way of AIs to govern AIs, seems the most

reliable path forward.

While the development of a humane (or humanistic) AI is our priority, we recognize some potential role for artificial humans. To the extent that we are able to develop individual self-engineering for specific abilities, thereby enabling some humans to match aspects of a future AI's intelligence, such a project could be useful. Of course, that effort would need to be the product of individual choice. Our authorial caution here is reflective of our collective dilemma: Evolution cannot merely be replaced by design, for this would be the forsaking of humanity. But ceding the project of discovery itself—whether spiritual, physical, scientific, or philosophical—would effectuate the same.

In the age of AI, the tension between the need to *design* ourselves and the need to *align* our creation may very well become our orienting compass. Both are as aspirational as they can also be conservative. It is yet to be determined to what extent the two needs will be viewed as fundamentally in contradiction with each other. If we give full play to AI's powers for unlimited exploration in this new age, we risk passivity or, worse, paralysis. But if we maximize our control, giving at least the illusion of safety, we circumscribe the extent of our fullest potential. Can we effectively mediate the exercise of our powers—our ever-advancing abilities of design and discovery—by reasserting a common and evolving conception of humanity?

We desire a future in which human intelligence and machine intelligence empower one another. To achieve that end, each intelligence must have an adequate understanding of the other. Defining who we are is just the first step, for to be human is not a constant. Far more work is needed to render our machines, and ourselves, transparent, legible, and reliable. Even if that goal is achieved for the snapshot of a moment, calibrating and sharing our truths and realities will be an intensive and ongoing project. In this way, the questions of coevolution and coexistence are not simply to be answered; they are to be enacted.